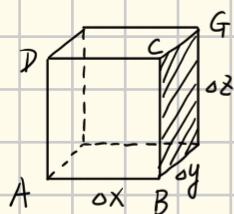


Chapter 7 平面势流和涡流

7.1 流体微团运动分析

1. 速度分解



$$U_G = u + \frac{\partial u}{\partial x} \Delta x + \frac{\partial u}{\partial y} \Delta y + \frac{\partial u}{\partial z} \Delta z$$

$$V_G = v + \frac{\partial v}{\partial x} \Delta x + \frac{\partial v}{\partial y} \Delta y + \frac{\partial v}{\partial z} \Delta z$$

$$W_G = w + \frac{\partial w}{\partial x} \Delta x + \frac{\partial w}{\partial y} \Delta y + \frac{\partial w}{\partial z} \Delta z$$

$$U_G \text{ 式可改写成: } U_G = u + \frac{\partial u}{\partial x} \Delta x + \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \Delta y + \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \Delta z + \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \Delta z - \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \Delta y$$

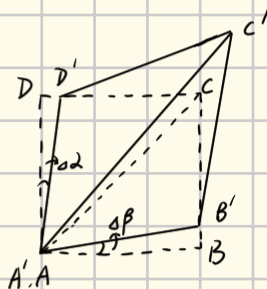
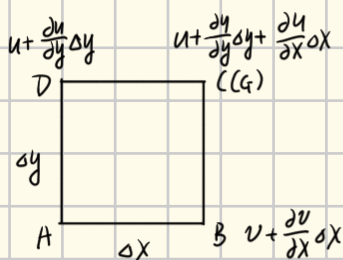
同理改写 V_G, W_G 引入

$$\begin{cases} \varepsilon_x = \frac{\partial u}{\partial x} & \varepsilon_y = \frac{\partial v}{\partial y} & \varepsilon_z = \frac{\partial w}{\partial z} \\ \gamma_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right) & \gamma_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) & \gamma_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \\ \omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) & \omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) & \omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \end{cases}$$

$$\text{则} \begin{cases} U_G = u + \varepsilon_x \Delta x + (\gamma_y \Delta z + \gamma_z \Delta y) + (\omega_y \Delta z - \omega_x \Delta y) \\ V_G = v + \varepsilon_y \Delta y + (\gamma_z \Delta x + \gamma_x \Delta z) + (\omega_z \Delta x - \omega_x \Delta z) \\ W_G = w + \varepsilon_z \Delta z + (\gamma_x \Delta y + \gamma_y \Delta x) + (\omega_x \Delta y - \omega_y \Delta x) \end{cases}$$

2. 线变形率

微团在 xoy 投影平面的变形



DC 在 x 方向上的投影长度会发生变化

假设 $\Delta t \propto \Delta x$, 则

$$\Delta l_x = \left(u + \frac{\partial u}{\partial x} \Delta x + \frac{\partial u}{\partial y} \Delta y \right) \Delta t - \left(u + \frac{\partial u}{\partial y} \Delta y \right) \Delta t = \frac{\partial u}{\partial x} \Delta x \Delta t$$

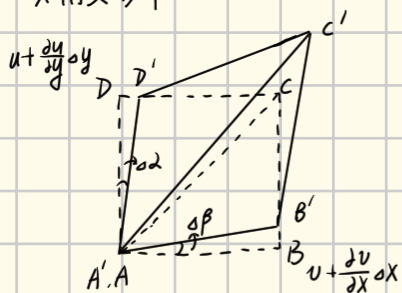
$$\lim_{\Delta t \rightarrow 0} \frac{\Delta l_x}{\Delta t} = \frac{\partial u}{\partial x} \Delta x = \varepsilon_x \Delta x$$

$$\text{定义 } \varepsilon = \varepsilon_x + \varepsilon_y + \varepsilon_z = \nabla \cdot \vec{v}$$

对 $\rho = \text{const}$, 由连续性方程有: $\varepsilon = \nabla \cdot \vec{v} = 0$

即要体不可压缩流体质量守恒, 则任意流体微团体积变形率必须为 0

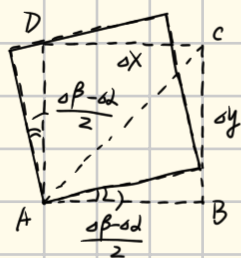
3. 角变形率



$$\alpha = \tan \alpha = \frac{(v + \frac{\partial v}{\partial x} dx) dt - v dt}{dx} = \frac{\partial v}{\partial x} dt$$

$$\alpha = \tan \alpha = \frac{(u + \frac{\partial u}{\partial y} dy) dt - u dt}{dy} = \frac{\partial u}{\partial y} dt$$

当 $\alpha \neq \alpha$ 时, 微团不仅有角变形, 还有整体的旋转



$$\frac{\alpha + \alpha}{dt} = (\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}) = 2\omega_z$$

在 XOy 平面内, G点 X 方向上的速度为 $y\omega_z$

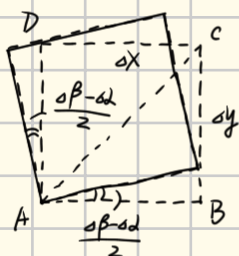
同理在 XOz 平面内, G点 X 方向上的速度为 $y\omega_z$

$\therefore$ G在 X 方向上产生的速度为 $y\omega_z + y\omega_z$

$$y \quad y\omega_z + y\omega_z$$

$$z \quad y\omega_z + y\omega_z$$

4. 旋转角速度



$$-\frac{\alpha - \alpha}{2dt} dy = -\frac{1}{2} (\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}) dy = -\omega_z dy$$

同样微团在 XOz 平面旋转在 X 方向的速度变化 $y\omega_z$

则 G点, 在 X 方向产生速度变化 $y\omega_z - y\omega_z$

$$y \quad y\omega_z - y\omega_z$$

$$z \quad y\omega_z - y\omega_z$$

$\omega_x, \omega_y, \omega_z$ 是微团单位时间内分别在 yOz, xOz, xOy 投影面内角速度 $\vec{\omega} = \omega_x \vec{i} + \omega_y \vec{j} + \omega_z \vec{k}$

$$\vec{\omega} = \frac{1}{2} (\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}) \vec{i} + \frac{1}{2} (\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}) \vec{j} + \frac{1}{2} (\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}) \vec{k} = \frac{1}{2} (\nabla \times \vec{v}) \quad \text{速度旋度}$$

$$\omega_r = \frac{1}{2} (\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} - \frac{\partial v_r}{\partial r}) \quad \omega_\theta = \frac{1}{2} (\frac{\partial v_r}{\partial r} - \frac{\partial v_\theta}{\partial \theta}) \quad \omega_z = \frac{1}{2} (\frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} - \frac{1}{r} \frac{\partial v_r}{\partial \theta})$$

5. 有旋、无旋

无旋: $\omega_x = \omega_y = \omega_z = 0$

$$\text{即} \begin{cases} \frac{\partial w}{\partial y} = \frac{\partial v}{\partial z} \\ \frac{\partial u}{\partial z} = \frac{\partial w}{\partial x} \\ \frac{\partial v}{\partial x} = \frac{\partial u}{\partial y} \end{cases}$$

7.2 速度环量与涡旋强度

1. 速度环量

$$\Gamma = \oint_L \vec{v} \cdot d\vec{s} = \oint_L u dx + v dy + w dz$$

2. 旋涡强度

$$\text{涡量 } \vec{\Omega} = \nabla \times \vec{v} = 2\vec{\omega}$$

$$\text{旋涡强度 } I = \int_A \vec{\Omega} \cdot \vec{n} dA = 2 \int_A \vec{\omega} \cdot \vec{n} dA = \int_A (\nabla \times \vec{v}) \cdot \vec{n} dA$$

3. 斯托克斯定理

$$\Gamma = \oint_L \vec{v} \cdot d\vec{s} = \int_A (\nabla \times \vec{v}) \cdot \vec{n} dA = \int_A \vec{\Omega} \cdot \vec{n} dA = I$$

7.3 旋涡运动的基本概念

$$1. \text{ 涡线 } \frac{dx}{\omega_x} = \frac{dy}{\omega_y} = \frac{dz}{\omega_z}$$

涡管 在旋涡场中通过一封闭曲线(非涡线)的每一点作涡线, 这些涡线形成一个管状表面。

涡索 截面积趋于无限小的涡管。

2. 开尔文定理

当理想、正压流体在有势质量力的作用下发生运动时, 沿任意封闭流体线的速度环量在运动过程中不随时间变化。

3. 旋涡运动的生成

流体具有粘性, 流体是非正压的及非有势质量力的作用是生成旋涡运动的原因。

7.4 不可压缩流体势流的基本求解方法

数学中的定理

设 G 是平面上的单连通区域: $P(x,y) \in G, Q(x,y) \in G$, 则

$\int P(x,y)dx + Q(x,y)dy$ 在 G 内与路径无关的充要条件:

$$\Leftrightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

$\Leftrightarrow P(x,y)dx + Q(x,y)dy$ 是某函数的全微分

即存在 $u(x,y)$ 满足 $du(x,y) = P(x,y)dx + Q(x,y)dy$.

扩展到三维问题

$P(x,y,z) \in G, Q(x,y,z) \in G, R(x,y,z) \in G, \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}, \frac{\partial P}{\partial z} = \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial z} = \frac{\partial R}{\partial y}$

$\Leftrightarrow \int P(x,y,z)dx + Q(x,y,z)dy + R(x,y,z)dz$ 与积分路径无关

$\Leftrightarrow \exists du = P(x,y,z)dx + Q(x,y,z)dy + R(x,y,z)dz$.

1. 速度势

势函数 $\varphi \Rightarrow d\varphi = udx + vdy + wdz$

全微分 $d\varphi = \frac{\partial \varphi}{\partial x} dx + \frac{\partial \varphi}{\partial y} dy + \frac{\partial \varphi}{\partial z} dz \Rightarrow \begin{cases} \frac{\partial \varphi}{\partial x} = u \\ \frac{\partial \varphi}{\partial y} = v \\ \frac{\partial \varphi}{\partial z} = w \end{cases}$

$\vec{v} = \text{grad } \varphi$

柱坐标下的势函数: $\begin{cases} v_r = \frac{\partial \varphi}{\partial r} \\ v_\theta = \frac{1}{r} \frac{\partial \varphi}{\partial \theta} \\ v_z = \frac{\partial \varphi}{\partial z} \end{cases}$

速度势的性质

(1) 调和函数

$$\Delta \varphi = \frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} = 0$$

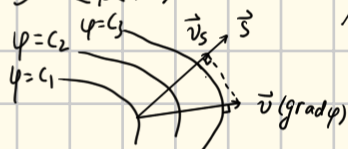
(2) 等势面(线)与流线垂直(正交)

(3) 速度沿任意曲线的线积分等于曲线两端速度势之差

$$\int_A^B \vec{v} \cdot d\vec{r} = \int_A^B udx + vdy + wdz = \int_A^B d\varphi = \varphi_B - \varphi_A$$

(4) 速度势在任意方向的方向导数等于速度在该方向的投影

$$\frac{\partial \varphi}{\partial s} = \text{grad } \varphi \cdot \vec{s} = v_s$$



2. 流函数

平面不可压流体 连续方程: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow \frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y}$

设 $d\psi = -vdx + udy$

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy \Rightarrow \begin{cases} \frac{\partial \psi}{\partial x} = -v \\ \frac{\partial \psi}{\partial y} = u \end{cases}$$

极坐标下的流函数

$$v_r = \frac{\partial \psi}{\partial \theta} \quad v_\theta = -\frac{\partial \psi}{\partial r}$$

流函数的性质

(1) 调和函数 $\Delta \psi = 0$

(2) 流函数的等值线是流线

(3) 两条流线间流量(单位厚度)等于两条流线流函数的差

$$Q = \int_A^B \vec{v} \cdot d\vec{r} = \int_A^B (udy - vdx) = \int_A^B d\psi = \psi_B - \psi_A$$

3. 势流函数的关系

(1) 两者同时存在的条件

势函数 — 无旋

流函数 — 平面流动、不可压缩

(2) 两者同时存在时, 都满足 Laplace 方程, 互为共轭函数.

$$\frac{\partial \varphi}{\partial x} = \frac{\partial \psi}{\partial y}, \quad \frac{\partial \varphi}{\partial y} = -\frac{\partial \psi}{\partial x}$$

(3) 势线与流函数等值线相互正交, 构成流网.

7.5 基本的平面有势流动

1. 均匀直线流

$$\varphi = u_0 x \quad \psi = u_0 y$$

2. 源与汇

$$\text{速度分布} \begin{cases} v_r = \pm \frac{Q}{2\pi r} & \text{"+" 源} \\ v_\theta = 0 & \text{"-" 汇} \end{cases}$$

$$\varphi = \pm \frac{Q}{2\pi} \ln r \quad \psi = \pm \frac{Q}{2\pi} \theta$$

3. 点涡

$$\text{速度分布} \begin{cases} v_r = 0 \\ v_\theta = \frac{\Gamma}{2\pi r} \end{cases} \quad \begin{aligned} \varphi &= \frac{\Gamma}{2\pi} \theta \\ \psi &= -\frac{\Gamma}{2\pi} \ln r \end{aligned}$$

4. 偶极

$$\lim_{\substack{Q \rightarrow 0 \\ Qr \rightarrow M}} Q \Delta r = M$$

↳ 偶极子强度

$$\varphi = \frac{M}{2\pi} \frac{x}{x^2+y^2} \quad \psi = -\frac{M}{2\pi} \frac{y}{x^2+y^2}$$

7.6 平面势流的叠加

1. 均匀绕圆柱体的无环量流动

均匀流 + 偶极

$$\text{边界条件: } v_r|_{r=0} = 0, \quad v_r|_{r=\infty} = u_\infty$$

$$\text{均匀流: } \psi_1 = u_\infty y \quad \text{设 } \psi_1 = u_\infty y$$

$$\text{偶极: } \psi_2 = -\frac{M}{2\pi} \frac{y}{x^2+y^2}$$

$$\text{叠加: } \psi = u_\infty y - \frac{M}{2\pi} \frac{y}{x^2+y^2}$$

$$\text{零流线 } \psi = 0 \quad \begin{cases} y=0 \\ u_\infty - \frac{M}{2\pi} \frac{1}{x^2+y^2} = 0 \Rightarrow x^2+y^2 = \frac{M}{2\pi u_\infty} \end{cases} \quad R = \sqrt{\frac{M}{2\pi u_\infty}} \Rightarrow M = 2\pi u_\infty R^2$$

$$\therefore \psi = V_{\infty} \left(y - \frac{R^2 y}{x^2 + y^2} \right) \quad \text{或} \quad \psi = V_{\infty} \left(r - \frac{R^2}{r} \right) \sin \theta.$$

$$v_r = \frac{\partial \psi}{r \partial \theta} = V_{\infty} \cos \theta \left(1 - \frac{R^2}{r^2} \right)$$

$$v_{\theta} = -\frac{\partial \psi}{\partial r} = -V_{\infty} \sin \theta \left(1 + \frac{R^2}{r^2} \right)$$

无穷远处: $v_r = V_{\infty} \cos \theta, v_{\theta} = -V_{\infty} \sin \theta$ (外边条)

柱面上: $r = R \quad \begin{cases} v_r = 0 \\ v_{\theta} = -2V_{\infty} \sin \theta \end{cases}$ (内边条)

柱面上速度特点: (1) $\theta = 90^\circ, 270^\circ, v = v_{\max}$ (驻点) $\theta = 0^\circ, 180^\circ$ (驻点) $v_r = 0, v_{\theta} = -2V_{\infty} \sin \theta.$

(2) 速度对称分布.

无旋场欧拉积分方程: $\frac{v^2}{2g} + z + \frac{P}{\rho g} = \frac{v_{\infty}^2}{2g} + z_{\infty} + \frac{P_{\infty}}{\rho g} \Rightarrow \frac{\rho v^2}{2} + P = \frac{\rho v_{\infty}^2}{2} + P_{\infty}$

柱面上的速度代入式得 $P - P_{\infty} = \frac{\rho v_{\infty}^2}{2} (1 - 4 \sin^2 \theta) \Rightarrow \frac{P - P_{\infty}}{\frac{1}{2} \rho v_{\infty}^2} = (1 - 4 \sin^2 \theta).$

令 $C_p = \frac{P - P_{\infty}}{\frac{1}{2} \rho v_{\infty}^2}$ 压强系数

Chapter 8 黏性不可压缩流体的运动

8.1 黏性流体中的应力

1. 黏性流体中的应力

一点的应力

$$P = \begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_{zz} \end{bmatrix}$$

切向应力互等定理

$$\left. \begin{aligned} \tau_{xy} &= \tau_{yx} \\ \tau_{yz} &= \tau_{zy} \\ \tau_{zx} &= \tau_{xz} \end{aligned} \right\}$$

2. 广义牛顿内摩擦定律

$$\begin{aligned} \sigma_{xx} &= -p + 2\mu \frac{\partial u}{\partial x} + \lambda \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) \\ \sigma_{yy} &= -p + 2\mu \frac{\partial v}{\partial y} + \lambda \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) \\ \sigma_{zz} &= -p + 2\mu \frac{\partial w}{\partial z} + \lambda \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) \end{aligned}$$

不可压缩

$$\begin{aligned} \sigma_{xx} &= -p + 2\mu \frac{\partial u}{\partial x} \\ \sigma_{yy} &= -p + 2\mu \frac{\partial v}{\partial y} \\ \sigma_{zz} &= -p + 2\mu \frac{\partial w}{\partial z} \end{aligned}$$

$$\begin{aligned} \tau_{xy} &= \tau_{yx} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \\ \tau_{yz} &= \tau_{zy} = \mu \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \\ \tau_{zx} &= \tau_{xz} = \mu \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) \end{aligned}$$

8.2 不可压缩黏性流体运动的基本方程

1. x方向

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) + f_x$$

2. y方向

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) + f_y$$

3. z方向

$$\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) + f_z$$

$$\Rightarrow \frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} = \vec{f} - \frac{1}{\rho} \nabla p + \nu \nabla^2 \vec{v}$$

特殊情况:

① 理想流体运动微分方程: $\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} = \vec{f} - \frac{1}{\rho} \nabla p$

② 静止流体或均匀流平衡方程: $\vec{f} = \frac{1}{\rho} \nabla p$

两平行平壁之间的定常平行流动

重力场: $f_x = 0, f_y = -g, f_z = 0$

水平层流: $v = w = 0$

速度与z无关: $\frac{\partial w}{\partial z} = 0$

定常流: $\frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} = \frac{\partial w}{\partial t} = 0$

连续性方程: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \Rightarrow \frac{\partial u}{\partial x} = 0 \Rightarrow \frac{\partial^2 u}{\partial x^2} = 0$

$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = f_x - \frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \Rightarrow \frac{\partial p}{\partial x} = \mu \frac{\partial^2 u}{\partial y^2}$ $\xrightarrow{u \text{ 是 } y \text{ 的单值函数}} \frac{\partial p}{\partial x} = \mu \frac{d^2 u}{dy^2}$

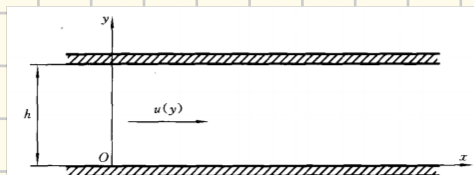


图 8-4 两平行平壁间的黏性流体流动

对 y 两次积分: $u = \frac{1}{2\mu} \frac{\partial P}{\partial x} y^2 + C_1 y + C_2$

1) 两板固定, 有压差

边界条件: $y=0, u=0$; $y=h, u=0$

$$\Rightarrow C_1 = -\frac{1}{2\mu} \frac{\partial P}{\partial x} h, C_2 = 0$$

$$\therefore u = \frac{1}{2\mu} \frac{\partial P}{\partial x} (y^2 - yh)$$

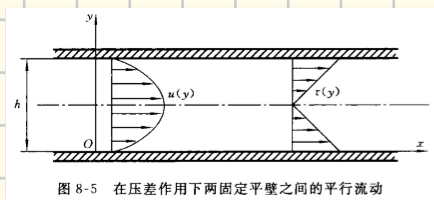


图 8-5 在压差作用下两固定平壁之间的平行流动

(2) 上板运动, 有压差

边界条件: $y=0, u=0$; $y=h, u=U$

$$\Rightarrow C_1 = \frac{U}{h} - \frac{1}{2\mu} \frac{\partial P}{\partial x} h, C_2 = 0$$

$$\therefore u = \frac{U}{h} y + \frac{1}{2\mu} \frac{\partial P}{\partial x} (y^2 - hy)$$

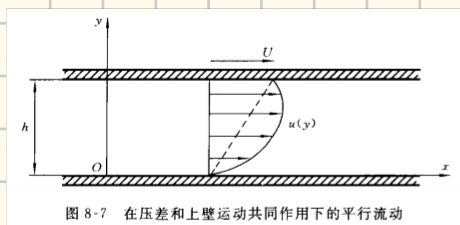


图 8-7 在压差和上壁运动共同作用下的平行流动

(3) 上板运动, 无压差

边界: $y=0, u=0$; $y=h, u=U$; $\frac{\partial P}{\partial x} = 0$

$$\Rightarrow C_1 = \frac{U}{h}, C_2 = 0$$

$$\therefore u = \frac{U}{h} y$$

8.4 边界层的基本概念及基本方程

1. 边界层概念

边界层: 实际流体绕过物体流动时, 由于流体粘性的影响, 在物体表面附近, 形成沿面的法线方向速度变化很快的薄层

边界层特点:

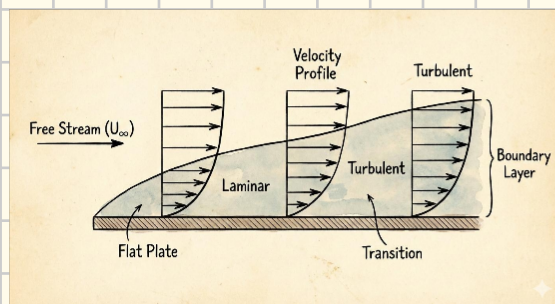
① $\delta \ll L$

② $\delta = \delta(x)$

③ 层内速度梯度 $\frac{du}{dy}$ 很大

④ 各截面上压强相等, 即 $\frac{\partial P}{\partial y} = 0$

⑤ 边界层粘性力与惯性力同量级



PS: ① 规定达到主流速度 99% 处为边界层边缘 $u|_{y=\delta} = 0.99U$

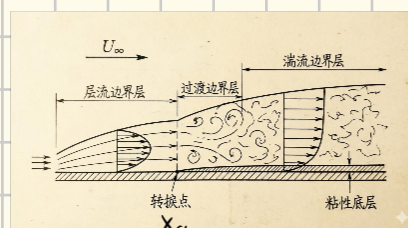
② 边界层外边界不是流线

边界层内流态

当地雷诺数: $Re_x = \frac{Ux}{\nu}$

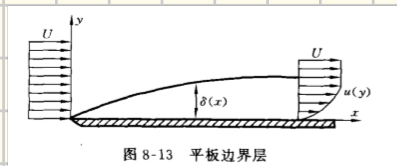
临界雷诺数: $Re_{cr} = \frac{Ux_{cr}}{\nu}$

对平板流动: $Re_{cr} = 5 \times 10^5 \sim 3 \times 10^6$



2. 边界层厚度、位移厚度、动量损失厚度

1) 边界层厚度



2) 位移厚度

$$\delta^* = \int_0^{\infty} \left(1 - \frac{u}{U}\right) dy$$

3) 动量损失厚度

$$\delta^{**} = \int_0^{\infty} \frac{u}{U} \left(1 - \frac{u}{U}\right) dy$$

3. 边界层流动基本方程

根据量级分析法, 由 N-S 方程导出.

$$\begin{cases} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \\ u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \frac{\partial^2 u}{\partial y^2} \\ \frac{\partial p}{\partial y} = 0 \end{cases} \quad \text{在边界层外边缘, } u=U, \frac{\partial u}{\partial y} = \frac{\partial v}{\partial y} = 0 \quad \Rightarrow \quad \begin{cases} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \\ u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = U \frac{dU}{dx} + \nu \frac{\partial^2 u}{\partial y^2} \end{cases}$$

边界: $y=0, u=v=0; y=\delta, u=U$

8.5 平板层流边界层的相似性解

$$\eta = y \sqrt{\frac{U}{2\nu x}} \quad \frac{u}{U} = f'(\eta) \quad f'f'' + f''' = 0$$

1. 速度分布 $\frac{u}{U} = f'(\eta)$

2. 边界层厚度 $\delta(x) = 5.0 \sqrt{\frac{\nu x}{U}} = 5.0 \frac{x}{\sqrt{Re_x}}$

3. 位移厚度 $\delta^*(x) = \sqrt{\frac{\nu x}{U}} \lim_{\eta \rightarrow \infty} [\eta - f(\eta)] \quad \delta^*(x) = 1.721 \sqrt{\frac{\nu x}{U}} = 1.721 \frac{x}{\sqrt{Re_x}}$

4. 动量损失厚度 $\delta^{**}(x) = \sqrt{\frac{\nu x}{U}} 2f''(0) = 0.664 \frac{x}{\sqrt{Re_x}}$

5. 壁面切应力及摩擦阻力系数

$$\tau_w = \mu U \sqrt{\frac{U}{\nu x}} f''(0) = 0.332 \frac{\rho U^2}{\sqrt{Re_x}}$$

$$F_D = \int_0^L \tau_w dx = 0.664 \frac{\rho U^2 L}{\sqrt{Re_x}}$$

$$C_f = \frac{F_D}{\frac{1}{2} \rho U^2 L} = \frac{1.328}{\sqrt{Re_x}}$$

8.6 边界层动量积分方程

$$\frac{d\delta^{**}}{dx} + \frac{\delta^{**}}{U} (2+H) \frac{dU}{dx} = \frac{\tau_w}{\rho U^2} \quad H = \frac{\delta^*}{\delta^{**}}$$

平板边界层: $\frac{d\delta^{**}}{dx} = \frac{\tau_w}{\rho U^2}$

8.7 湍流边界层与混合边界层

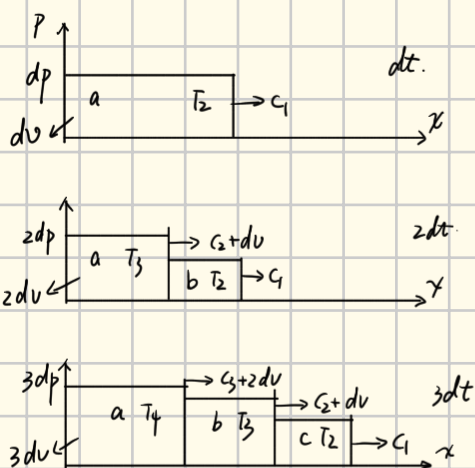
$$\frac{u}{U} = \left(\frac{y}{\delta}\right)^{1/7} \quad \frac{\tau_w}{\rho U^2} = 0.0225 \left(\frac{\nu}{U\delta}\right)^{1/4}$$

Chapter 9 激波

激波：气流参数发生突跃变化的压缩波。

激波分类：正激波、斜激波、曲激波。

正激波的形成



正激波基本关系式

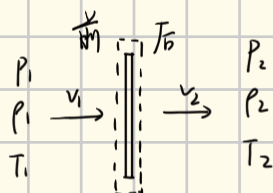
基本方程：

连续方程 $\rho_1 u_1 = \rho_2 u_2$

动量方程 $p_1 - p_2 = \rho_2 u_2^2 - \rho_1 u_1^2$

能量方程 $h_1 + \frac{1}{2} u_1^2 = h_2 + \frac{1}{2} u_2^2$

状态方程 $\frac{p_1}{\rho_1 T_1} = \frac{p_2}{\rho_2 T_2}$



普朗特激波公式： $u_1 u_2 = c_*^2$ 或 $\lambda_1 \lambda_2 = 1$

结论：① 波前超声，波后亚声

② u_1 越大， u_2 越小； $\lambda_1 \rightarrow 1$ ， $\lambda_2 \rightarrow 1$

Rankine-Hugoniot 公式

$$\frac{p_2}{p_1} = \frac{(\gamma+1) \frac{p_2}{p_1} - (\gamma-1)}{(\gamma+1) - (\gamma-1) \frac{p_2}{p_1}}$$

或

$$\frac{p_2}{p_1} = \frac{(\gamma+1) \frac{p_2}{p_1} + (\gamma-1)}{(\gamma+1) + (\gamma-1) \frac{p_2}{p_1}}$$

或

$$\frac{T_2}{T_1} = \frac{\frac{\gamma+1}{\gamma-1} \frac{p_2}{p_1} + \left(\frac{p_2}{p_1}\right)^2}{\frac{\gamma+1}{\gamma-1} \frac{p_2}{p_1} + 1}$$

$$\frac{p_2}{p_1} \rightarrow \infty \Rightarrow \frac{p_2}{p_1} \rightarrow \frac{\gamma+1}{\gamma-1} \quad \text{空气 } \gamma=1.4 \Rightarrow \frac{\gamma+1}{\gamma-1} = 6$$

结论：激波是突跃压缩过程，是不可逆绝热过程，压缩有限。

(1) 上下游马赫数关系

(2) 密度比与上游马赫数关系

$$Ma_2^2 = \frac{2 + (\gamma+1) Ma_1^2}{2\gamma Ma_1^2 - (\gamma+1)}$$

$$\frac{\rho_2}{\rho_1} = \frac{u_1}{u_2} = \frac{u_1^2}{u_1 u_2} = \frac{u_1^2}{c_*^2} = \lambda_1^2 = \frac{(\gamma+1) Ma_1^2}{2 + (\gamma-1) Ma_1^2}$$

(3) 压强比与上游马赫数关系

$$\frac{p_2}{p_1} = \frac{2\gamma}{\gamma+1} Ma_1^2 - \frac{\gamma-1}{\gamma+1}$$

(4) 滞止压强与上游马赫数关系

$$\frac{p_{01}}{p_{02}} = \left(\frac{2\gamma}{\gamma+1} Ma_1^2 - \frac{\gamma-1}{\gamma+1} \right)^{\frac{1}{\gamma-1}} \left[\frac{2 + (\gamma-1) Ma_1^2}{(\gamma+1) Ma_1^2} \right]^{\frac{\gamma}{\gamma-1}}$$

(5) 熵增与上游马赫数关系

$$s_2 - s_1 = R \ln \frac{p_{01}}{p_{02}}$$

当 $Ma_1 > 1$ 时, 总有 $\frac{p_{01}}{p_{02}} > 1$, $s_2 > s_1$
激波压缩总是熵增过程.

(6) 临界有效喉部面积与上下游马赫数关系.

$$\frac{A_2^*}{A_1^*} = \frac{Ma_2}{Ma_1} \left[\frac{2 + (\gamma - 1) Ma_1^2}{2 + (\gamma - 1) Ma_2^2} \right]^{\frac{(\gamma - 1)(\gamma + 1)}{2}}$$

$$A_2^* > A_1^*$$